

Question:
A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?

- Answer Choices:
- A. Epstein-Barr virus (EBV)
 - B. Measles virus
 - C. Human immunodeficiency virus (HIV)
 - D. Human papillomavirus

Reasoning + Answer:

<think> The cluster of differentiation 21 cd 2 1 is a receptor for the complement component 3 on the B cells and is essential for the b cells to be able to bind to the virus during the infection of the measles virus and take up the virus by endocytosis and present the viral antigens to t cells so removing cd 2 1 from the B cells will affect the measles virus infection and the binding of the virus to the B cells will be prevented </think>

<answer> B . Me as les virus </answer>

